

● HARD

● PROBLEM-SOLVING AND DATA ANALYSIS

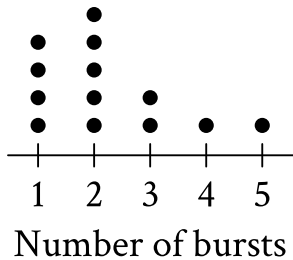
One-variable data: Distributions and measures of center and spread

03

3 questions · ~5 min

Q1

PROBLEM-SOLVING AND DATA ANALYSIS



- The data for the dot plot are as follows:
 - 1: 4 dots
 - 2: 5 dots
 - 3: 2 dots
 - 4: 1 dot
 - 5: 1 dot

The dot plot represents a data set of the number of bursts for 13 eruptions of a steam vent. If an additional eruption with 11 bursts is added to this data set to create a new data set of 14 eruptions, which of the following measures will be greater for the new data set than for the original data set?

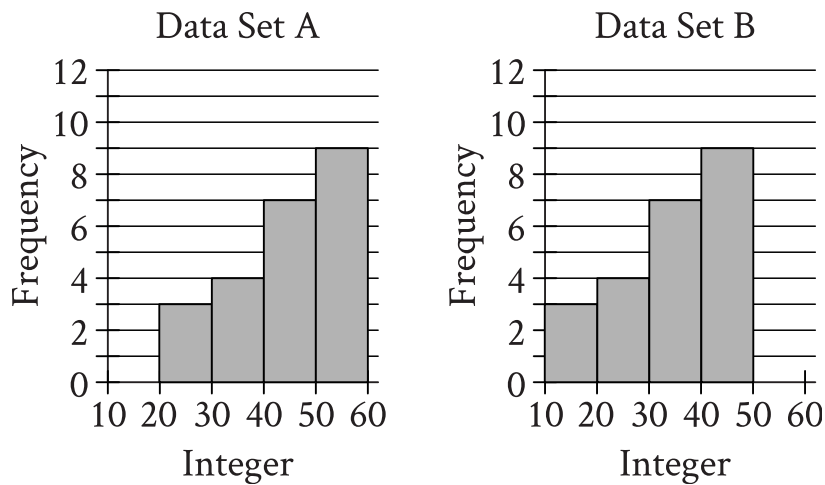
- I. The median number of bursts
 - II. The mean number of bursts
- A. I and II
- B. I only
- C. II only
- D. Neither I nor II

The table shown summarizes the number of employees at each of the 17 restaurants in a town.

Number of employees	Number of restaurants
2 to 7	2
8 to 13	4
14 to 19	2
20 to 25	7
26 to 31	2

Which of the following could be the median number of employees for the restaurants in this town?

- A. 2
- B. 9
- C. 15
- D. 21



- Data Set A:
 - The horizontal axis is labeled Integer. It ranges from 10 to 60 and is divided into 5 equal intervals.
 - The vertical axis is labeled Frequency. It ranges from 0 to 12 in increments of 1, with values marked every 2 grid lines.
 - The histogram has a skewed right shape.
 - The histogram has 4 bins.
 - The Frequency data for the 4 bins are as follows:
 - 20 to 30: 3
 - 30 to 40: 4
 - 40 to 50: 7
 - 50 to 60: 9
- Data Set B:
 - The horizontal axis is labeled Integer. It ranges from 10 to 60 and is divided into 5 equal intervals.
 - The vertical axis is labeled Frequency. It ranges from 0 to 12 in increments of 1, with values marked every 2 grid lines.
 - The histogram has a skewed right shape.
 - The histogram has 4 bins.
 - The Frequency data for the 4 bins are as follows:
 - 10 to 20: 3
 - 20 to 30: 4

- 30 to 40: 7
- 40 to 50: 9

Two data sets of 23 integers each are summarized in the histograms shown. For each of the histograms, the first interval represents the frequency of integers greater than or equal to 10, but less than 20. The second interval represents the frequency of integers greater than or equal to 20, but less than 30, and so on. What is the smallest possible difference between the mean of data set A and the mean of data set B?

- A. 0
- B. 1
- C. 10
- D. 23